

DO NOT WRITE ON TEST



**COMPUTER PROGRAMMING
CONCEPTS
~OPEN EVENT~
(390)**

STATE 2024

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Which type of network has a topology in which each node is connected to two other nodes to form a single network communication channel in which all of the nodes handle the transmission of the data?
 - A. Star Network
 - B. Bus Network
 - C. Tree Network
 - D. Ring Network
2. Which of the following terms represents the coexistence of telephone, video, and data transmissions on a network?
 - A. Star Network
 - B. Network Convergence
 - C. Network Data Management Protocol
 - D. Tree Network
3. What is the “.jar” file type?
 - A. Java archive file
 - B. Java application run file
 - C. Java application run bytecode
 - D. Java archive source code
4. What is the JVM?
 - A. Java Virtual Machine
 - B. JavaScript Virtual Machine
 - C. J++ Virtual Machine
 - D. Java Vector Math for game development
5. Which statement best defines Machine Code?
 - A. Code that is written in 1's and 0's
 - B. A language that is designed for ease of use
 - C. Code that is designed to be close to human language
 - D. A language with a high level of abstraction
6. If your computer is unable to connect to the internet, which port below would be the best choice to check for a bad connection?
 - A. RJ45
 - B. USB
 - C. VGA
 - D. HDD

7. What does GPU stand for?
 - A. Graphics Processing Unit
 - B. Gateway Power Unit
 - C. Graphics Power Unit
 - D. Gateway Processing Unit
8. Which of the following best describes how Unicode differs from ASCII?
 - A. Unicode encodes special texts from different languages, letters, and symbols
 - B. Unicode encodes digits, letters, and symbols
 - C. Unicode is a subset of ASCII
 - D. Unicode encodes only non-English symbols
9. The base-10 number system used by computer is also known as which of the following?
 - A. Octal
 - B. Hexadecimal
 - C. Decimal
 - D. Binary
10. This type of application software represents data in rows and columns, in addition you can perform calculations on this data:
 - A. Tabulation Software
 - B. Spreadsheet Software
 - C. Data Science Software
 - D. Database Software
11. In an OOP based language what is the strategy of basing an object upon another object and keeping its behaviors and attributes?
 - A. Inheritance
 - B. Template
 - C. Encapsulation
 - D. Abstraction
12. What is needed to declare a class in Python?
 - A. Class keyword
 - B. Class heading and set of curly braces { } for data members
 - C. Class heading, set of curly braces { } for data members, and a constructor
 - D. Class heading, set of curly braces { } for data members, and data attributes
13. In Java, what is trait of a data attribute that is declared private?
 - A. The data can be written only from any outside class
 - B. The data can be read only from any outside class
 - C. The data can be read/write from any outside class
 - D. Data attributes cannot be accessed from any outside class

14. Which type of data structure is best defined as each node had two children nodes?
 - A. Binary Trees
 - B. Stacks
 - C. Array List
 - D. Queues
15. In a Binary Tree data structure, what is the name of the topmost data node?
 - A. Super Class
 - B. Parent
 - C. Root
 - D. Parent Class
16. Which type of data structure is best defined as linear data structure where the data elements follow an order where operations are performed on them either by LIFO or FILO?
 - A. Linked List
 - B. Stack
 - C. Array
 - D. Queue
17. Which type of Heap data structure is best defined as having a key at the root that is the least value compared to all children nodes?
 - A. Super Min Heap
 - B. Linear Min Heap
 - C. Minimum Heap
 - D. Min Heap
18. What is one of the advantages of using OOP?
 - A. Uses data abstraction
 - B. Uses procedure abstraction
 - C. Follows a top down approach
 - D. Program is divided into small parts called functions
19. Which of the following would be considered a hierarchical data structure?
 - A. Binary Search Tree
 - B. Binary Array Tree
 - C. Binary Linked List
 - D. All of the above
20. Which of the following is the best reason for a programmer to use a binary tree data structure?
 - A. When the program is working data elements that are transferred asynchronously
 - B. When the program needs to search through elements or objects that are organized in a hierarchy
 - C. When the program needs to perform insertion and deletions much faster
 - D. When the program needs to search through homogenous elements stored in contiguous locations

21. Which type of algorithm strategy uses an approach that breaks the problem into smaller parts by calling the same function until a certain condition is met?
- A. Hashing
 - B. Brute Force
 - C. Backtracking
 - D. Recursive
22. Which type of algorithm strategy breaks the problem into smaller problems, solves the simpler problem and merges them to find the final solution?
- A. Hashing
 - B. Divide and Conquer
 - C. Dynamic Programming
 - D. Sorting
23. Which term below best matched the following definition: a set of rules for solving a mathematical problem in a set number of steps that sometimes uses recursion.
- A. Bubble Sorting
 - B. Data Traversal
 - C. Recursive Heap
 - D. Algorithm
24. Which of the following should be the first step in Structured Design?
- A. Perform a needs assessment
 - B. Evaluate the system
 - C. Generate alternatives
 - D. None of these
25. What type of loop has an initialization, condition, and increments/decrements all in a single line?
- A. While
 - B. For
 - C. Foreach
 - D. Do While
26. What is an advantage of a static array?
- A. Efficient execution
 - B. Creates excess space in case it needs to grow
 - C. Can expand and contract as needed
 - D. Slow to compile but at run time it can have its speed increased upon user input
27. Which logical operator would best match with the following statement: if X is true execute, else do not.
- A. AND Operator
 - B. OR Operator
 - C. NOT Operator
 - D. IF-ELSE

28. In the Java class `Java.lang.System`, which field best matches the following definition: this stream is open and ready to supply input data; works directly with keyboard input or another form of input chosen by the user.
- A. `public static final InputStream in`
 - B. `public static final StringStream in`
 - C. `public static final CharStream in`
 - D. `public static final PrintStream in`
29. In Java, which exception is involved with a scenario where copying, accessing, or sorting a data element is outside of the parameters of the range in which it resides?
- A. `ElementOutOfBoundsException`
 - B. `ArrayOutOfBoundsException`
 - C. `RangeOutOfBoundsException`
 - D. `IndexOutOfBoundsException`
30. In Java, which of the following are compile time exceptions?
- A. `NullPointerException`
 - B. `IOException`
 - C. `ClassCastException`
 - D. All of the above
31. In Java, which of the following are run time exceptions?
- A. `NullPointerException`
 - B. `ArrayIndexOutOfBoundsException`
 - C. `ClassCastException`
 - D. All of the above
32. Which of the following are part of the Java Exception Hierarchy?
- A. `Throwable`
 - B. `Error`
 - C. `Exception`
 - D. All of the above
33. Which of the following best matches the feature of having a fixed size at compile time?
- A. Dynamic array
 - B. Static array
 - C. Static method
 - D. Dynamic method
34. In Java, which of the following is NOT a compile time exception?
- A. `DateTimeException`
 - B. `InstantiationException`
 - C. `InterruptedException`
 - D. `SQLException`

35. Which type of data structure is best defined as a collection of items of the same type stored in a continual section of memory, so it is very easy to discover the location of each element?
- A. Binary Trees
 - B. Stacks
 - C. Array
 - D. Queues
36. In which stage of the software development cycle would market research be performed?
- A. Designing
 - B. Identification
 - C. Marketing
 - D. Market Analysis
37. What do software engineers use for loads, verifies and executes Java bytecode?
- A. JAR
 - B. JRE
 - C. JVM
 - D. JDK
38. What is the name of the structural view of ULM?
- A. Static
 - B. Behavioral
 - C. Dynamic
 - D. All of the above
39. What shape in a flowchart indicates the beginning or ending of a process?
- A. Arrow
 - B. Diamond/ Rhombus
 - C. Oval
 - D. Parallelogram
40. What is the most common usage of SQL?
- A. Accessing, communicating and altering with a database
 - B. Managing the user forms on a website
 - C. Providing data analytics from web scraping
 - D. None of these

41. What is printed from the code below:

```
double x = 11.9;
double y = 6.9;
char[][] bpa = new char[(int)Math.round(y)][(int)Math.floor(x)];
bpa[1] = new char[] {1,2,'A',3,'D'};
System.out.println( bpa[1].length + bpa[2].length );
```

- A. 12
- B. 14
- C. 16
- D. 18

42. What is printed from the code below:

```
String one = "This is a string";
String two = new String("This is a string");
if(((one = two) == one) == false)
    System.out.println("It worked!");
else if (one != two)
    System.out.println("Nevermind");
else
    System.out.println(one == two);
```

- A. It worked!
- B. Nevermind
- C. True
- D. False

43. What is printed from the code below:

```
String x = "BPA is the best!";
x+= "What happened?";
int y = x.length()-(x.length()-4);
char z = x.charAt(y);
System.out.println(x.charAt(z-(z-2)));
```

- A. B
- B. P
- C. A
- D. t

44. What is printed from the code below:

```
boolean v = true;  
boolean x = false;  
int y = 7, z = 8;  
System.out.println( (x ^ v) + " y+z: " + y + z );
```

- A. true y+z: 15
- B. false y+z: 15
- C. true y+z: 78
- D. false y+z: 78

45. Convert the base-2 number “111101” to a base-10

- A. 15
- B. 30
- C. 61
- D. 123

46. Convert the base-10 number “14” to a base-16 (hex)

- A. C
- B. D
- C. E
- D. F

47. What is printed from the code below:

```
System.out.print((int)4.5 / 4.0 * (double)2 + 2.5 + (int)1.5);
```

- A. 4
- B. 5
- C. 4.5
- D. 5.5

48. What is printed from the code below:

```
int A = 'A';  
char B = 'A';  
System.out.print( A+B );
```

- A. AA
- B. 0
- C. AB
- D. 130

49. Choose the SQL script below that will allow the program to give how many “cars” are in each “category” in the “currentInventory” table and display the number of cars in each category:
- A. SELECT category, count(category) FROM currentInventory GROUP BY category;
 - B. SELECT cars, count(cars) FROM currentInventory GROUP BY category;
 - C. SELECT category, count(cars) FROM currentInventory GROUP BY cars;
 - D. SELECT category, add(cars) FROM currentInventory GROUP BY category(cars);
50. Choose the SQL script below that will allow the program to give the user the update MODEL column and set the value “F150” in all rows where the Type is “Truck”
- A. MODIFY Vehicles SET MODEL = ‘F150’ WHERE Type = ‘Truck’;
 - B. UPDATE Vehicles SET Type = ‘Truck’ WHERE Type = ‘F150’;
 - C. SELECT Vehicles SET MODEL = ‘F150’ WHERE Type = ‘Truck’;
 - D. UPDATE Vehicles SET MODEL = ‘F150’ WHERE Type = ‘Truck’;